

EC-341 Digital System Design

BS(CE)-2021

LAB REPORT # 3



Submitted By:

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

**Lab Instructor.
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Experiment #3

Overview of ZYBO

Objective:

1. Overview of ZYBO.
2. Zynq AP SoC Architecture.
3. What is Constraints File?
4. Lab Tasks.

Software Used:

Xilinx Vivado

Procedure:

The ZYBO (ZYNq BOard) is a feature-rich, ready-to-use, entry-level embedded software and digital circuit development platform built around the smallest member of the Xilinx Zynq-7000 family, the Z-7010. The Z-7010 is based on the Xilinx All Programmable System-on-Chip (AP SoC) architecture, which tightly integrates a dual-core ARM Cortex-A9 processor with Xilinx 7-series Field Programmable Gate Array (FPGA) logic. When coupled with the rich set of multimedia and connectivity peripherals available on the ZYBO, the Zynq Z-7010 can host a whole system design. The on-board memories, video and audio I/O, dual-role USB, Ethernet, and SD slot will have your design up-and-ready with no additional hardware needed. Additionally, six Pmod connectors are available to put any design on an easy growth path. The Zynq 7010 AP SoC offers the following features:

1. 650Mhz dual-core Cortex-A9 processor
2. DDR3 memory controller with 8 DMA channels
3. High-bandwidth peripheral controllers: 1G Ethernet, USB 2.0, SDIO
4. Low-bandwidth peripheral controller: SPI, UART, CAN, I2C
5. Reprogrammable logic equivalent to Artix-7 FPGA
 - i. 4,400 logic slices, each with four 6-input LUTs and 8 flip-flops
 - ii. 240 KB of fast block RAM
 - iii. Two clock management tiles, each with a phase-locked loop (PLL) and mixed-mode clock manager (MMCM)
 - iv. 80 DSP slices
 - v. Internal clock speeds exceeding 450MHz
 - vi. On-chip analog-to-digital converter (XADC)

Tasks

Task 1:

Write a program which toggles LED using a slide switch.

CODE:

```
23 module LED(  
24     input switch,  
25     input led  
26 );  
27     assign led=switch;  
28 endmodule
```

```

8 set_property -dict { PACKAGE_PIN L16 IOSTANDARD LVCMOS33 } [get_ports { clk }]; #IO_L11P_T1_SRCC_35 Sch=sysclk
9 create_clock -add -name sys_clk_pin -period 8.00 -waveform {0 4} [get_ports { clk }];
10
11
12 ###Switches
13 set_property -dict { PACKAGE_PIN G15 IOSTANDARD LVCMOS33 } [get_ports { switch1 }]; #IO_L19N_T3_VREF_35 Sch=SW0
14 #set_property -dict { PACKAGE_PIN P15 IOSTANDARD LVCMOS33 } [get_ports { switch2 }]; #IO_L24P_T3_34 Sch=SW1
15 #set_property -dict { PACKAGE_PIN W13 IOSTANDARD LVCMOS33 } [get_ports { b[0] }]; #IO_L4N_T0_34 Sch=SW2
16 #set_property -dict { PACKAGE_PIN T16 IOSTANDARD LVCMOS33 } [get_ports { b[1] }]; #IO_L9P_T1_DQS_34 Sch=SW3
17
18
19 ###Buttons
20 set_property -dict { PACKAGE_PIN R18 IOSTANDARD LVCMOS33 } [get_ports { button1 }]; #IO_L20N_T3_34 Sch=BTN0
21 #set_property -dict { PACKAGE_PIN P16 IOSTANDARD LVCMOS33 } [get_ports { BTN1 }]; #IO_L24N_T3_34 Sch=BTN1
22 #set_property -dict { PACKAGE_PIN V16 IOSTANDARD LVCMOS33 } [get_ports { BTN2 }]; #IO_L18P_T2_34 Sch=BTN2
23 #set_property -dict { PACKAGE_PIN Y16 IOSTANDARD LVCMOS33 } [get_ports { BTN3 }]; #IO_L7P_T1_34 Sch=BTN3
24
25
26 ###LEDs
27 set_property -dict { PACKAGE_PIN M14 IOSTANDARD LVCMOS33 } [get_ports { led1 }]; #IO_L23P_T3_35 Sch=LED0
28 #set_property -dict { PACKAGE_PIN M15 IOSTANDARD LVCMOS33 } [get_ports { led2 }]; #IO_L23N_T3_35 Sch=LED1
29 #set_property -dict { PACKAGE_PIN C14 IOSTANDARD LVCMOS33 } [get_ports { led3 }]; #IO_L23P_T3_35 Sch=LED2

```

Output:



Conclusion:

In this lab we had design the zybo, by using its hardware connect into the Xilinx Vivado and also we check the working Zybo.